

- 1.3 (a) First of all, $\mathcal{F}_6$ satisfies the first two conditions of being a σ -algebra. Secondly, for any $\{A_n\}_{n \geq 1} \subset \mathcal{F}_6$, if $\cup_{n \geq 1} A_n$ is countable then $\cup_{n \geq 1} A_n \in \mathcal{F}_6$. If $\cup_{n \geq 1} A_n$ is not countable, then $\exists$ uncountable set $A_{n_0} \in \{A_n\}_{n \geq 1}$. Because $A_{n_0} \in \mathcal{F}_6$, so $A_{n_0}^c$ has to be countable, and consequently $A_{n_0}^c \supset \cap_{n \geq 1} A_n^c = (\cup_{n \geq 1} A_n)^c$ is countable, i.e. $\cup_{n \geq 1} A_n \in \mathcal{F}_6$.
- (b) As we can see, we only need to verify the *countable additivity* property for μ on $\mathcal{F}_6$. For any disjoint collection $\{A_n\}_{n \geq 1} \subset \mathcal{F}_6$, by the argument in (a), we see that there can be at most one uncountable set in $\{A_n\}_{n \geq 1}$, because they are disjoint ($\therefore \cup_{n \neq j} A_n \subset A_j^c, \forall j$). Thus $\mu(\cup_{n \geq 1} A_n) = \sum_{n \geq 1} \mu(A_n)$, and $(\Omega, \mathcal{F}_6, \mu)$ is a measure space.

1.8 Clearly, $\sigma\langle \mathcal{A} \rangle \subset \mathcal{P}(\mathbb{N})$.

Secondly, $\forall$ nonempty set $J \in \mathcal{P}(\mathbb{N})$, $J = \cup_{i \in J} \{i\} = \cup_{i \in J} (A_i \cap A_{i+1}^c) \in \sigma\langle \mathcal{A} \rangle$ (Notice that $J \subset \mathbb{N}$ is always countable).

Thus, $\sigma\langle \mathcal{A} \rangle = \mathcal{P}(\mathbb{N})$.

1.10 To simplify this prove, we introduce several easy facts (try to convince yourself that they are true):

1) $\Omega_1 \subset \Omega_2 \Rightarrow \sigma\langle \Omega_1 \rangle \subset \sigma\langle \Omega_2 \rangle$.

2) $\sigma\langle \mathcal{F} \rangle = \mathcal{F}, \forall \sigma$ -algebra $\mathcal{F}$.

3)

$$\begin{aligned} \bigcup_{i \geq 1} (A \times B_i) &= A \times \left(\bigcup_{i \geq 1} B_i \right), & \bigcup_{i \geq 1} (B_i \times A) &= \left(\bigcup_{i \geq 1} B_i \right) \times A \\ (A \times B) \setminus (A \times C) &= A \times (B \setminus C), & (B \times A) \setminus (C \times A) &= (B \setminus C) \times A \end{aligned}$$

$$\bigcap_{i \geq 1} \bigcap_{j \geq 1} (A_i \times B_j) = \left(\bigcap_{i \geq 1} A_i \right) \times \left(\bigcap_{j \geq 1} B_j \right).$$

$$(A \times B)^c = (A^c \times B^c) \cup (A^c \times B) \cup (A \times B^c)$$

4) For any $(a, b) \subset \mathbb{R}$, $(a, b) = \cup_{n \geq 1} [(-\infty, b) \setminus (-\infty, a + n^{-1})]$.

In **Example 1.1.6**, it has been proved that

$$\mathcal{B}(\mathbb{R}^k) = \sigma\langle \mathcal{O}_1 \rangle = \sigma\langle \mathcal{O}_3 \rangle,$$

so it suffices to prove $\mathcal{O}_3 \subset \sigma\langle \mathcal{O}_4 \rangle$ in order to get $\mathcal{B}(\mathbb{R}^k) = \sigma\langle \mathcal{O}_1 \rangle = \sigma\langle \mathcal{O}_2 \rangle = \sigma\langle \mathcal{O}_3 \rangle = \sigma\langle \mathcal{O}_4 \rangle$ and consequently $\mathcal{O}_j \subset \sigma\langle \mathcal{O}_i \rangle, \forall 1 \leq i, j \leq 4$. This is because $\mathcal{O}_4 \subset \mathcal{O}_2 \Rightarrow \sigma\langle \mathcal{O}_4 \rangle \subset \sigma\langle \mathcal{O}_2 \rangle \subset \mathcal{B}(\mathbb{R}^k)$, if $\mathcal{O}_3 \subset \sigma\langle \mathcal{O}_4 \rangle$, then $\sigma\langle \mathcal{O}_3 \rangle \subset \sigma\langle \mathcal{O}_4 \rangle$.

For simplicity, we introduce the following notation

$$I(\alpha_1, \dots, \alpha_m) = (-\infty, \alpha_1) \times \dots \times (-\infty, \alpha_m)$$

$\forall (a_1, b_1) \times \dots \times (a_k, b_k) \in \mathcal{O}_3, a_i, b_i \in \mathcal{Q}, a_i \leq b_i, 1 \leq i \leq k$, let

$$A_i = I(b_1, \dots, b_{i-1}) \times (a_i, b_i) \times I(b_{i+1}, \dots, b_k), 1 \leq i \leq k,$$

especially, $A_1 = (a_1, b_1) \times I(b_2, \dots, b_k)$, $A_k = I(b_1, \dots, b_{k-1}) \times (a_k, b_k)$. Using fact 3) and 4), it can be seen that $\bigcap_{1 \leq i \leq k} A_i = (a_1, b_1) \times \dots \times (a_k, b_k)$, and

$$\begin{aligned} A_i &= I(b_1, \dots, b_{i-1}) \times \bigcup_{n \geq 1} [(-\infty, b_i) \setminus (-\infty, a_i + n^{-1})] \times I(b_{i+1}, \dots, b_k) \\ &= \bigcup_{n \geq 1} \{I(b_1, \dots, b_{i-1}) \times [(-\infty, b_i) \setminus (-\infty, a_i + n^{-1})] \times I(b_{i+1}, \dots, b_k)\} \\ &= \bigcup_{n \geq 1} \{I(b_1, \dots, b_k) \setminus I(b_1, \dots, b_{i-1}, a_i + n^{-1}, b_{i+1}, \dots, b_k)\} \\ &\in \sigma\langle \mathcal{O}_4 \rangle, \end{aligned}$$

So $\bigcap_{1 \leq i \leq k} A_i \in \sigma\langle \mathcal{O}_4 \rangle$, consequently, $\mathcal{O}_3 \subset \sigma\langle \mathcal{O}_4 \rangle$.

Remark:

To give a better idea about how this works, let's take a look at the case when $k = 2$. In this case, $\forall (a_1, b_1) \times (a_2, b_2) \in \mathcal{O}_3$,

$$\begin{aligned} (a_1, b_1) \times (-\infty, b_2) &= \bigcup_{n \geq 1} \{(-\infty, b_1) \times (-\infty, b_2) \setminus (-\infty, a_1 + n^{-1}) \times (-\infty, b_2)\} \\ (-\infty, b_1) \times (a_2, b_2) &= \bigcup_{n \geq 1} \{(-\infty, b_1) \times (-\infty, b_2) \setminus (-\infty, b_1) \times (-\infty, a_2 + n^{-1})\}, \end{aligned}$$

which means that both $(a_1, b_1) \times (-\infty, b_2)$ and $(-\infty, b_1) \times (a_2, b_2)$ are in $\sigma\langle \mathcal{O}_4 \rangle$, and consequently their intersection $(a_1, b_1) \times (a_2, b_2) \in \sigma\langle \mathcal{O}_4 \rangle$.

1.18 This problem gives us the idea of the necessariness, in *m.c.f.a.*, of $\mu(A_m) < \infty$ for some m .

$$\lim_{n \rightarrow \infty} \mu(A_n) = \infty \neq 0 = \mu(\emptyset) = \mu(\bigcap_{n \geq 1} A_n)$$

1.19 (a) By definition of algebra, $\mathcal{A}(\mathcal{C}) \subset \mathcal{E}$, for any algebra $\mathcal{E} \supset \mathcal{C}$. So we only need to show that $\mathcal{A}(\mathcal{C})$ is an algebra.

Pick any $A \in \mathcal{C}$, by definition of semi-algebra, $\exists B_1, \dots, B_m \in \mathcal{C}$ such that $A^c = \bigcup_{i=1}^m B_i$. Now $\Omega = A \cup A^c = A \cup \{\bigcup_{i=1}^m B_i\} \in \mathcal{A}(\mathcal{C})$. Note that Ω may not be in the semi-algebra $\mathcal{C}$, e.g. $\Omega = \{1, 2, 3\}, \mathcal{C} = \{\emptyset, \{1\}, \{2, 3\}\}$.

Secondly, $\mathcal{A}(\mathcal{C})$ is closed under finite unions, by definition of $\mathcal{A}(\mathcal{C})$.

Further, for any set $A \in \mathcal{A}(\mathcal{C})$, $A = \bigcup_{i=1}^k B_i, B_i \in \mathcal{C}, i = 1, \dots, k$. Similarly, $B_i^c = \bigcup_{j=1}^{n_i} B_{ij}$. Then $A^c = \bigcap_{i=1}^k B_i^c = \bigcap_{i=1}^k \{\bigcup_{j=1}^{n_i} B_{ij}\}$, or further that

$$A^c = \bigcap_{i=1}^k \left\{ \bigcup_{j=1}^{n_i} B_{ij} \right\} = \bigcup_{j_1=1}^{n_1} \dots \bigcup_{j_k=1}^{n_k} \left(\bigcap_{m=1}^k B_{mj_m} \right) \in \mathcal{A}(\mathcal{C})$$

The above fact is true because semi-algebra is closed under finite intersections. So $\mathcal{A}(\mathcal{C})$ is the algebra generated by $\mathcal{C}$.

- (b) $\mathcal{C} \subset \mathcal{A}(\mathcal{C}) \Rightarrow \sigma\langle\mathcal{C}\rangle \subset \sigma\langle\mathcal{A}(\mathcal{C})\rangle$, $\mathcal{A}(\mathcal{C}) \subset \sigma\langle\mathcal{C}\rangle \Rightarrow \sigma\langle\mathcal{A}(\mathcal{C})\rangle \subset \sigma\langle\sigma\langle\mathcal{C}\rangle\rangle = \sigma\langle\mathcal{C}\rangle$. So $\sigma\langle\mathcal{C}\rangle = \sigma\langle\mathcal{A}(\mathcal{C})\rangle$.

1.20 As we mentioned earlier, Ω may not be in its corresponding semi-algebra $\mathcal{C}$, but $\mathcal{C}$ will have $\emptyset$ for sure, which can be easily seen using definition of semi-algebra (convince yourself). Check the definition of the *outer measure* using the following notation:

$$\Sigma(A) = \left\{ \sum_{n=1}^{\infty} \mu(A_n) : \{A_n\}_{n \geq 1} \subset \mathcal{C}, A \subset \bigcup_{n \geq 1} A_n \right\}, \quad \mu^*(A) = \inf \Sigma(A)$$

- (a) $0 \leq \mu^*(\emptyset) \leq \sum_{n \geq 1} \mu(\emptyset) = 0 \Rightarrow \mu^*(\emptyset) = 0$.
(b) $A \subset B \Rightarrow \Sigma(B) \subset \Sigma(A) \Rightarrow \mu^*(A) = \inf \Sigma(A) \leq \inf \Sigma(B) = \mu^*(B)$.
(c) By definition of infimum, we know that $\forall \epsilon > 0$, for each $i, \exists \{A_{ij}\}_{j \geq 1} \subset \mathcal{C}, A_i \subset \bigcup_{j \geq 1} A_{ij}$, such that $\sum_{j \geq 1} \mu(A_{ij}) \leq \mu^*(A_i) + \frac{\epsilon}{2^i}$. Thus $\sum_{i \geq 1} \sum_{j \geq 1} \mu(A_{ij}) \leq \sum_{i \geq 1} \mu^*(A_i) + \epsilon$. This means that $\bigcup_{i \geq 1} A_i \subset \bigcup_{i \geq 1} \bigcup_{j \geq 1} A_{ij}$, $\{A_{ij} : i \geq 1, j \geq 1\} \subset \mathcal{C}$ and consequently $\mu^*(\bigcup_{i \geq 1} A_i) \leq \sum_{i \geq 1} \sum_{j \geq 1} \mu(A_{ij}) \leq \sum_{i \geq 1} \mu^*(A_i) + \epsilon, \forall \epsilon > 0 \Rightarrow \mu^*(\bigcup_{i \geq 1} A_i) \leq \sum_{i \geq 1} \mu^*(A_i)$.

1.24 We denote

$$\begin{aligned} \mathcal{C} &= \{(a, b] : -\infty \leq a \leq b < \infty\} \cup \{(a, \infty) : -\infty \leq a < \infty\} \\ \mathcal{C}_2 &= \{I_1 \times I_2 : I_1, I_2 \in \mathcal{C}\} \end{aligned}$$

- (i) $\forall I_1 \times I_2, I_3 \times I_4 \in \mathcal{C}_2, I_1 \times I_2 \cap I_3 \times I_4 = (I_1 \cap I_3) \times (I_2 \cap I_4) \in \mathcal{C}_2$, because $\mathcal{C}$ is a semi-algebra, who's closed under finite intersections (try to self-convince $\mathcal{C}$ is truly a semi-algebra).
(ii) $\forall I_1 \times I_2 \in \mathcal{C}_2, (I_1 \times I_2)^c = (I_1^c \times I_2) \cup (I_1 \times I_2^c) \cup (I_1^c \times I_2^c)$. Again, $I_k \in \mathcal{C}, k = 1, 2 \Rightarrow I_k^c = \bigcup_{j=1}^{n_k} I_{kj}, I_{kj_1} \cap I_{kj_2} = \emptyset, j_1 \neq j_2, I_{k1}, \dots, I_{kn_k} \in \mathcal{C}$. With the fact that $I_1 \times I_2^c, I_1^c \times I_2, I_1^c \times I_2^c$ are disjoint sets, $I_{kj} \cap I_{kl} = \emptyset, j \neq l, k = 1, 2$ and

$$\begin{aligned} I_1^c \times I_2 &= \bigcup_{j=1}^{n_1} (I_{1j} \times I_2) \\ I_1 \times I_2^c &= \bigcup_{j=1}^{n_2} (I_1 \times I_{2j}) \\ I_1^c \times I_2^c &= \bigcup_{j=1}^{n_1} \bigcup_{l=1}^{n_2} (I_{1j} \times I_{2l}) \end{aligned}$$

we see that $(I_1 \times I_2)^c$ is essentially a finite union of disjoint sets from $\mathcal{C}_2$, i.e. $(I_1 \times I_2)^c \in \mathcal{C}_2$.

- 1.x (i) Given any $x_1 \leq x_2$, after discussion of different cases of x_1, x_2 , it's always true that $F(x_1) \leq F(x_2)$. First, $F(x)$ is continuous at points other than 0, 1, and $\epsilon > 0, 0 = F(-\epsilon) = F(0) = \lim_{\epsilon \rightarrow 0+} \epsilon = \lim_{\epsilon \rightarrow 0+} F(\epsilon), 1 = F(1 + \epsilon) = F(1) = \lim_{\epsilon \rightarrow 0+} F(1 - \epsilon)$, thus $F(x)$ is continuous.
(ii) Use the semi-algebra $\mathcal{C}$ in (3.7), and define μ_F as in (3.8).
(iii) $\forall b \in \mathbb{R}, a \in \mathbb{R} \cup \{\pm\infty\}, (-\infty, b) = \bigcup_{n \geq 1} (-\infty, b - 1/n] \Rightarrow \mathcal{O}_2 \subset \sigma\langle\mathcal{C}\rangle \Rightarrow \mathcal{B}(\mathbb{R}) = \sigma\langle\mathcal{O}_2\rangle \subset \sigma\langle\mathcal{C}\rangle$. $(a, b] = \bigcap_{n \geq 1} (a, b + 1/n) \Rightarrow \mathcal{C} \subset \sigma\langle\mathcal{O}_1\rangle \Rightarrow \sigma\langle\mathcal{C}\rangle \subset \mathcal{B}(\mathbb{R})$. So $\sigma\langle\mathcal{C}\rangle = \mathcal{B}(\mathbb{R})$.

- (iv),(v) By *Caratheodory's extension theorem* and *uniqueness of measures*, there exists an unique measure μ on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ extending μ_F .

The logic may look like this, *Caratheodory's extension theorem* says, $\because \mu_F$ is a measure on semi-algebra $\mathcal{C}$, $\therefore \exists \mu_F^*$, induced ((3.1)) by the measure μ_F on semi-algebra $\mathcal{C}$, such that $(\mathbb{R}, \mathcal{M}_{\mu_F^*}, \mu_F^*)$ is a complete measure space, where $\mathcal{C} \subset \mathcal{M}_{\mu_F^*}$ is a σ -algebra, and $\mu_F^*(A) = \mu_F(A), \forall A \in \mathcal{C}$. Thus $\mathcal{B}(\mathbb{R}) = \sigma\langle \mathcal{C} \rangle \subset \mathcal{M}_{\mu_F^*}$. Now define measure μ as μ_F^* restricted on $\mathcal{B}(\mathbb{R})$, then $(\mathbb{R}, \mathcal{B}(\mathbb{R}), \mu)$ is a measure space (which may not be complete anymore). This extension is unique, because semi-algebra is always a π -system, by *uniqueness of measures* (Theorem 1.2.4), any two *finite* measures μ, μ' that equal to each other on a π -system $\mathcal{C}$, would be exactly the same one on a bigger collection, the $\sigma\langle \mathcal{C} \rangle$ (Note that the measure μ is a finite measure, actually a probability measure). The “common name” is uniform (0,1) measure on the real line.